Year 11 Set 4 · Foundation · Lesson 1

Percentage change & reverse percentages

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Today I will learn to...

Goals

1. Work out a percentage increase or decrease

2. Find the original amount (reverse percentages)

Do now: retrieval practice

4 mins

1. Find 3/4 of 60

   45

2. Write 0.35 as a percentage

   35%

3. Work out 7 × 0.4

   2.8

4. Round 3.476 to 1 dp

   3.5

Objective 1

Work out a percentage increase or decrease

Key idea

Learn

!

Percentage change = change ÷ original × 100. Always divide by the ORIGINAL amount.

Objective 1: Work out a percentage increase or decrease

I do: example 1

I do

A price goes from £40 to £46. Find the percentage increase.

1. Change = 46 − 40 = 6
2. 6 ÷ 40 = 0.15
3. 0.15 × 100 = 15% increase

I do: example 2

I do

A dog's mass falls from 80 kg to 68 kg. Find the percentage decrease.

1. Change = 80 − 68 = 12
2. 12 ÷ 80 = 0.15
3. 15% decrease

I do: example 3

I do

A car bought for £12 000 is sold for £9 000. Find the percentage loss.

1. Loss = 12 000 − 9 000 = 3 000
2. 3 000 ÷ 12 000 = 0.25
3. 25% loss

Mini whiteboards

We do

1. £50 → £60. Percentage increase?

Answer: 20%

2. 250 → 200. Percentage decrease?

Answer: 20%

3. £80 → £92. Percentage increase?

Answer: 15%

4. £120 → £90. Percentage decrease?

Answer: 25%

You do: choose your challenge

You do

Bronze

A number goes from 20 to 25. Find the percentage increase.

Silver

A TV price falls from £450 to £405. Find the percentage decrease.

Gold

Ali's pay rose from £24 000 to £25 200. Bea's rose from £30 000 to £31 350. Who had the bigger percentage rise?

Platinum

A shop buys a jacket for £35 and sells it for £56. Find the percentage profit. In a sale the selling price is cut by 25%. What is the percentage profit now?

You do: answers

Check

Bronze

25%

Silver

10%

Gold

Ali: 5%. Bea: 4.5%. Ali had the bigger rise

Platinum

60% profit. Sale price £42, so 20% profit

If you finish early

Extra

Bronze

A price goes from £40 to £50. Find the percentage increase.

25%

Silver

A phone drops from £300 to £255. Find the percentage decrease.

15%

Gold

A plant grows from 1.25 m to 1.6 m. Find the percentage increase.

0.35 ÷ 1.25 = 28%

Platinum

A shop buys a toy for £8 and sells it for £11. In a sale the price is cut by 20%. Find the percentage profit in the sale.

Sale price £8.80, profit 80p = 10%

Exam question 1

2 marks

Ali's rent rises from £800 to £860 a month. Work out the percentage increase.

Space for working

Mark scheme: 60 ÷ 800 × 100 = 7.5%

Exam question 2

2 marks

A shop sold 120 coats in October and 90 in November. Work out the percentage decrease.

Space for working

Mark scheme: 30 ÷ 120 × 100 = 25%

Objective 1 achieved

Work out a percentage increase or decrease



Objective 2

Find the original amount (reverse percentages)

✓ Achieved: Work out a percentage increase or decrease

Key idea

Learn

!

Write the new amount as a percentage of the original, turn it into a multiplier, then DIVIDE. 20% off → 80% → ÷ 0.8.

Objective 2: Find the original amount (reverse percentages)

I do: example 1

I do

In a 20% sale a coat costs £64. Find the original price.

1. After 20% off you pay 80%
2. Multiplier = 0.8
3. 64 ÷ 0.8 = £80

I do: example 2

I do

After a 15% increase a price is £92. Find the original price.

1. 115% → multiplier 1.15

2. 92 ÷ 1.15 = £80

I do: example 3

I do

A town's population fell by 10% to 4 500. What was it before?

1. 90% → multiplier 0.9

2. 4 500 ÷ 0.9 = 5 000

Mini whiteboards

We do

1. After 25% off a top costs £45. Original price?

Answer: £60

2. After a 10% rise a ticket costs £66. Original price?

Answer: £60

3. After 20% off a coat costs £64. Original price?

Answer: £80

4. After a 15% rise a bill is £92. Original amount?

Answer: £80

You do: choose your challenge

You do

Bronze

After 10% off, a book costs £27. Find the original price.

Silver

After a 25% increase, a price is £150. Find the original price.

Gold

A house fell in value by 4% to £216 000. What was it worth before?

Platinum

A price rises by 20% and then falls by 10%. It is now £540. Find the original price.

You do: answers

Check

Bronze	£30
Silver	£120
Gold	£225 000
Platinum	1.2 × 0.9 = 1.08, 540 ÷ 1.08 = £500

If you finish early

Extra

Bronze

After 50% off, shoes cost £22. Find the original price.

£44

Silver

After a 5% pay rise Sam earns £21 000. What did he earn before?

£20 000

Gold

A car lost 35% of its value and is now worth £9 100. What was it worth before?

9 100 ÷ 0.65 = £14 000

Platinum

20% VAT is added to a price. The total is £150. How much of the total is VAT?

150 ÷ 1.2 = £125, so VAT = £25

Exam question 1

3 marks

Normal prices are reduced by 30% in a sale. A jacket costs £42 in the sale. Work out the normal price.

Space for working

Mark scheme: 42 ÷ 0.7 = £60

Exam question 2

3 marks

Sam's wage goes up by 5% to £315 a week. Work out his wage before the rise.

Space for working

Mark scheme: 315 ÷ 1.05 = £300

Objective 2 achieved

Find the original amount (reverse percentages)



Real life

Apply

Tia buys a phone for £600 and sells it a year later for £420. What is her percentage loss?

180 ÷ 600 × 100 = 30% loss

Extension: stretch yourself

Stretch

1. A price rises by 20%, then falls by 20%. What is the overall percentage change?

$1.2 \times 0.8 = 0.96 \rightarrow 4\%$ decrease

2. A price including 20% VAT is £150. Find the price before VAT.

$150 \div 1.2 = £125$

Exit ticket: exam questions

Exit

1. A bike costs £240. Its price increases by 15%. Work out the new price. **(2 marks)**

2. A car's value falls from £8 000 to £6 800. Work out the percentage decrease. **(2 marks)**

3. In a sale, prices are reduced by 35%. A sofa costs £520 in the sale. Work out its normal price. **(3 marks)**

Exit ticket: mark scheme

Mark it

Q1 240 × 1.15 = £276 [2]

Q2 1 200 ÷ 8 000 × 100 = 15% [2]

Q3 520 ÷ 0.65 = £800 [3]

Total: _____ / 7

Today I achieved

- ✔ **Work out a percentage increase or decrease**
- ✔ **Find the original amount (reverse percentages)**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Increase £65 by 12%. [2]

2. The population of a village went from 1 250 to 1 400. Work out the percentage increase. [2]

3. Jo bought a bracelet for £24 and sold it for £18. Work out her percentage loss. [2]

4. After a 40% reduction, a coat costs £54. Work out the original price. [3]

5. A savings account adds 3% interest. The balance is now £1 545. How much was in the account before? [3]

Extra practice: Maths Genie → Edexcel question bank → Percentages · Reverse Percentages